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PDA Research

2019 Sterile Lyophilized Drug Product Loading Survey

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Parenteral Drug Association
Connecting People, Science and Regulation®

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PDA Survey
2019 Sterile Lyophilized Drug Product Loading Survey

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Introduction

The *2019 Sterile Lyophilized Drug Product Loading Survey* was designed to align and expand PDA's insight on current practices for companies who manufacture sterile lyophilized drug products, with particular focus on how firms conduct lyophilizer loading and protect the loading area from contamination by personnel.

Many firms cannot reduce or eliminate personnel from the lyophilizer loading area due to current designs. In these cases, learning of the methods employed to minimize the impact of personnel on loading is important. For other firms, with new facilities and state of the art technology either already implemented or soon to be implemented, understanding the regulatory influences and compliance considerations driving decision-making in the near term were also edifying.

The results of this survey were discusssed at the Packaging Science Interest Group Meeting during the PDA/FDA Conference, September 16-18, 2019, in Washington, DC. The full results will support future PDA initiatives pertaining to lyophilization and clean room operations, as appropriate.

Methodology

The *2019 Sterile Lyophilized Drug Product Loading Survey* was open to the general public for participation from June 25 to July 31, 2019.

The results are based on the views of **91** industry professionals across North America, Latin America, Europe, Asia/Pacific, Africa, and the Middle East. Survey respondents are involved in lyophilizer activities within their current companies who possess an understanding of their companies' procedures and needs.

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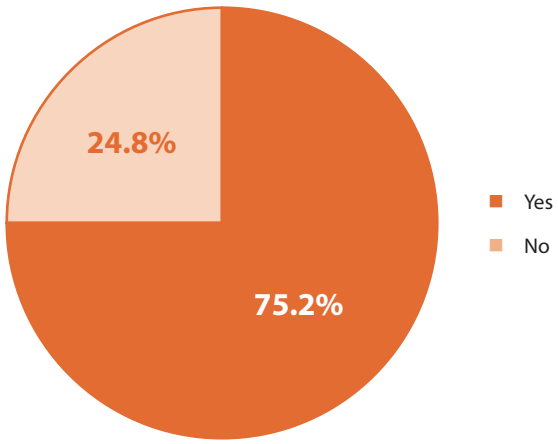
DEMOGRAPHICS (Questions 1-5)

Many organizations manufacture, develop, or contract manufacture lyophilized drug products. The initial survey questions queried the demographics of respondents to assess representation of the pharmaceutical and biopharmaceutical industries. The responses to the following questions demonstrate that the survey represents a broad span of industry with a good distribution of small and large companies, multiple product types, and global reach **(Questions 1–5).** Those who responded no to the first question did not answer any of the subsequent questions and the survey was ended.

1. Do you currently manufacture sterile lyophilized drug product?

Answer	Count	%
Yes	91	75.2%
No	30	24.8%
	Answered	***121***
	Skipped	***0***



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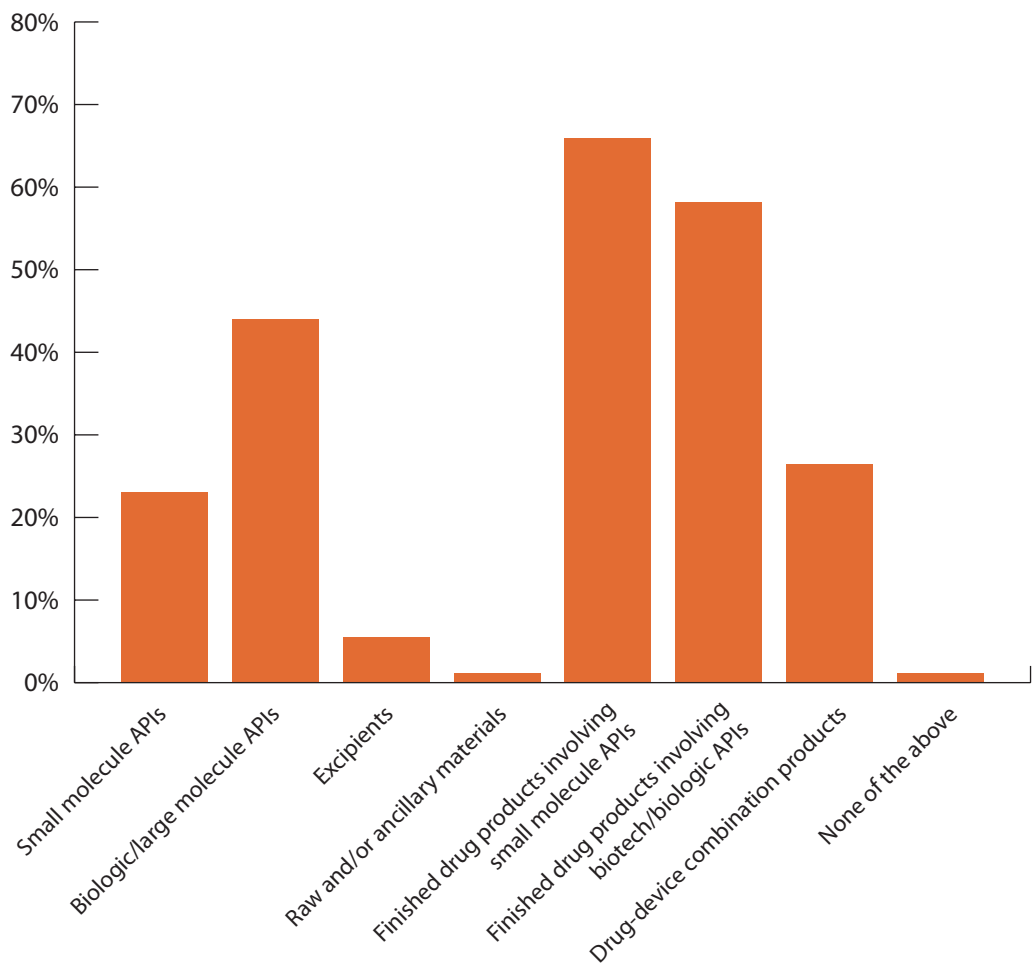
2. What types of products does your company manufacture, develop, or contract for manufacturing? (please check all that apply)

Answer	Count	%
Small molecule APIs	21	23.1%
Biologic/large molecule APIs	40	44.0%
Excipients	5	5.5%
Raw and/or ancillary materials	1	1.1%
Finished drug products involving small molecule APIs	60	65.9%
Finished drug products involving biotech/biologic APIs	53	58.2%
Drug-device combination products	24	26.4%
None of the above	1	1.1%
	Answered	***91***
	Skipped	***0***



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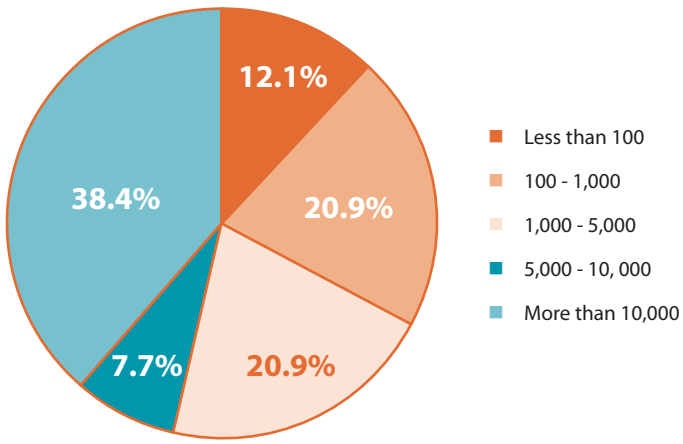
3. What is the size of your company (number of employees)?

Answer	Count	%
Less than 100	11	12.1%
100 - 1,000	19	20.9%
1,000 - 5,000	19	20.9%
5,000 - 10, 000	7	7.7%
More than 10,000	35	38.4%
	Answered	***91***
	Skipped	***0***



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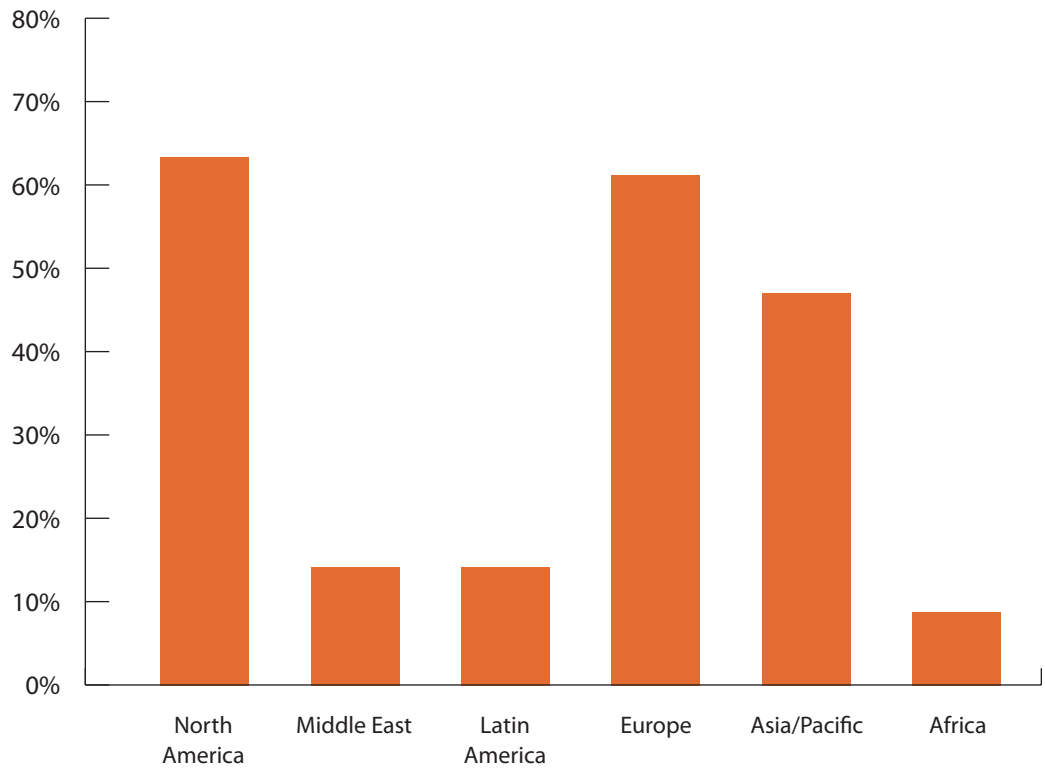
4. Where are your company's internal development/manufacturing locations? (please check all that apply)

Answer	Count	%
North America	58	63.7%
Middle East	13	14.3%
Latin America	13	14.3%
Europe	56	61.5%
Asia/Pacific	43	47.3%
Africa	8	8.8%
	Answered	***91***
	Skipped	***0***



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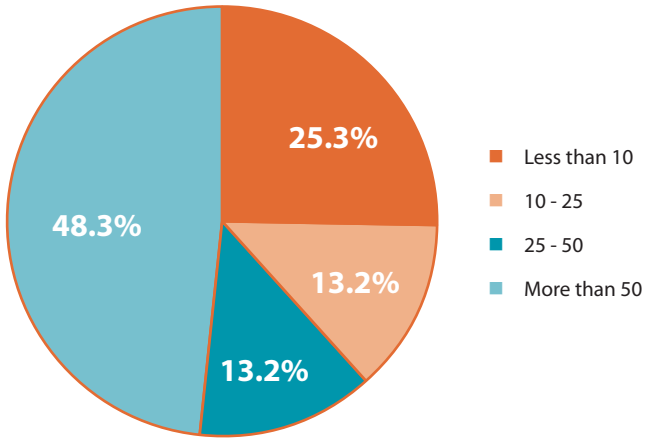
5. In how many markets are your products licensed or used in further manufacturing?

Answer	Count	%
Less than 10	23	25.3%
10-25	12	13.2%
25-50	12	13.2%
More than 50	44	48.3%
	Answered	***91***
	Skipped	***0***



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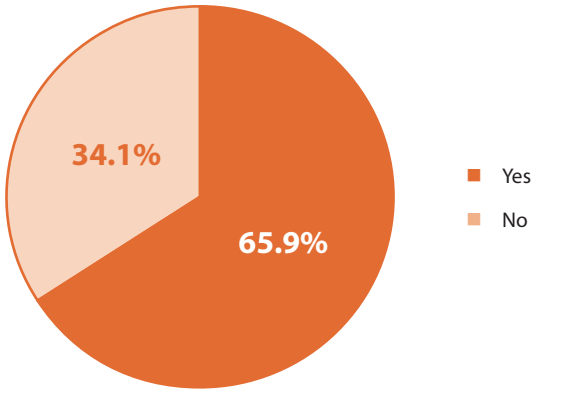
LYOPHILIZED PRODUCT LOADING (Questions 6–19)

For the survey respondents, it was important to establish what proportion of automated loading was in use and what proportion of their total loading operations that this represented. The following questions and responses would indicate this information **(Questions 6–9).**

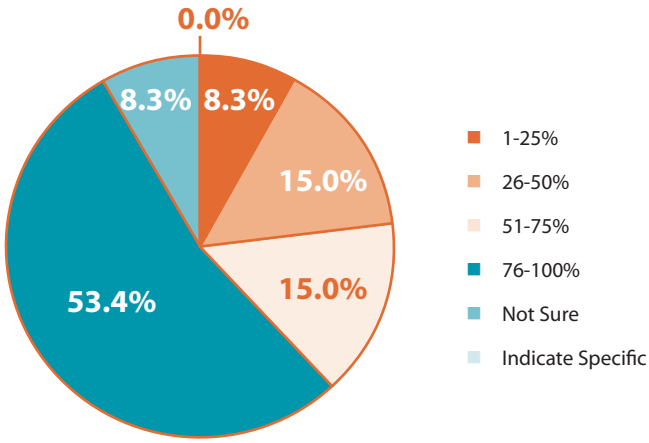
6. Do you have any completely automated systems for lyophilizer loading (e.g., no or limited manual/human participation during loading)?

Answer	Count	%
Yes	60	65.9%
No	31	34.1%
	Answered	***91***
	Skipped	***0***



7. What percent of all lyophilizer loading areas are fully automated?

Answer	Count	%
1-25%	5	8.3%
26-50%	9	15.0%
51-75%	9	15.0%
76-100%	32	53.4%
Not sure	5	8.3%
Indicate specific %	0	0.0%
	Answered	***60***
	Skipped	***31***



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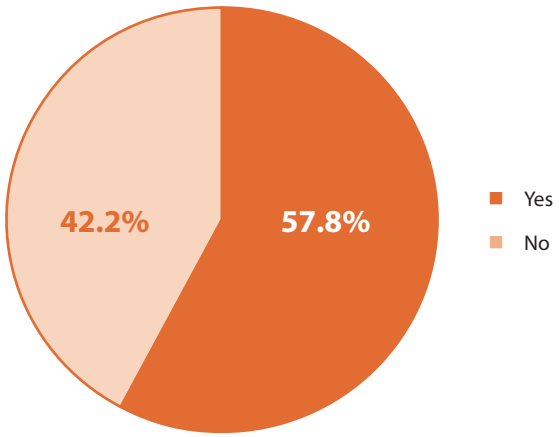
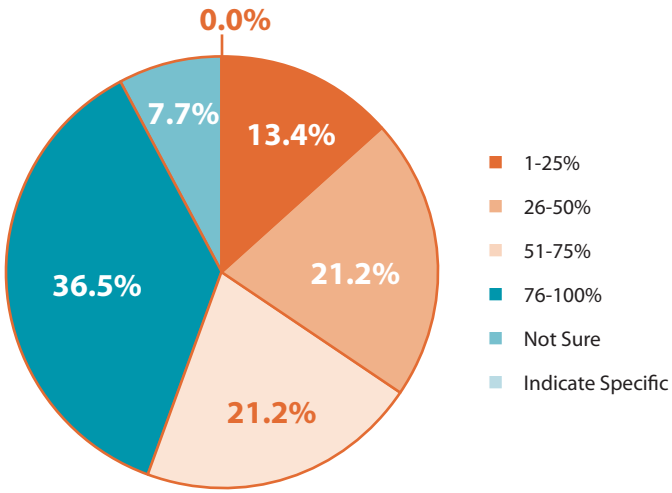
8. Do you have any manual processes for lyophilizer loading, where personnel are required in the Grade A/B space?

Answer	Count	%
Yes	52	57.8%
No	38	42.2%
	Answered	***90***
	Skipped	***1***



9. What percent of all lyophilizer loading areas are manual loading processes?

Answer	Count	%
1-25%	7	13.4%
26-50%	11	21.2%
51-75%	11	21.2%
76-100%	19	36.5%
Not sure	4	7.7%
Indicate specific %	0	0.0%
	Answered	***52***
	Skipped	***39***



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For operations with manual interventions, we next wanted to characterize the types of observations that might have been provided by regulatory agencies either through inspections or through the submission process. A number of questions were also posed to identify current risk mitigation measures as well as internal attitudes about changes to facilities and operations in the future **(Questions 10–19).**

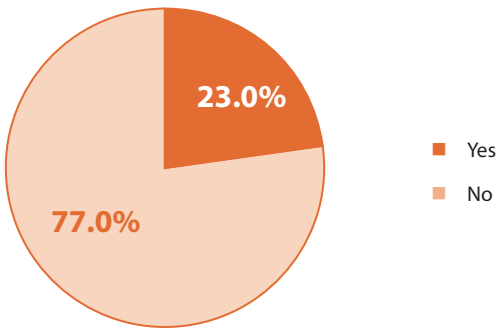
10. If your current lyophilizer processing area requires manual intervention, has your company received any Health Authority observations for the presence of operators in your lyophilizer processing area?

Answer	Count	%
Yes	20	23.0%
No	67	77.0%
	Answered	***87***
	Skipped	***4***

*If yes, please generalize the finding.

Responses:

- Additional Environmental sampling required during loading of Lyo. Sleeve monitoring should be increased.
- Appropriate classification of areas; Line design to remove people.
- Aseptic behavior not appropriate.
- Inadequate controls to minimize exposure of partially stoppered vials during loading of transport carts from filling line, and during loading/unloading from freeze dryers.
- It is difficult to generalize the findings.
- Manual interaction during loading and unloading of half stoppered vials into lyophilizer. Obstruction of air flow.
- Manual loading is not in line with regulatory expectations.
- Manual loading; presence of personnel in Grade A space.
- Personnel are too close to open product during loading. Airflow is at risk of flowing over operator onto the product.
- Risk of personnel manually loading / unloading, technology is not state of the art…
- Show the record obtained.
- That gowning should be with criteria of non-contamination, zero.
- The questions are: What type and interventions are performed in the filling manual? In which area classification is the filling done? In which area classification is the lyophilizer located? What type of cleaning is applied to freezer dryer? Is the lyophilisate process validated both in media fill and qualification process?
- They don't like manual loading operations and requested all products to their market be transferred to the automated loading line.



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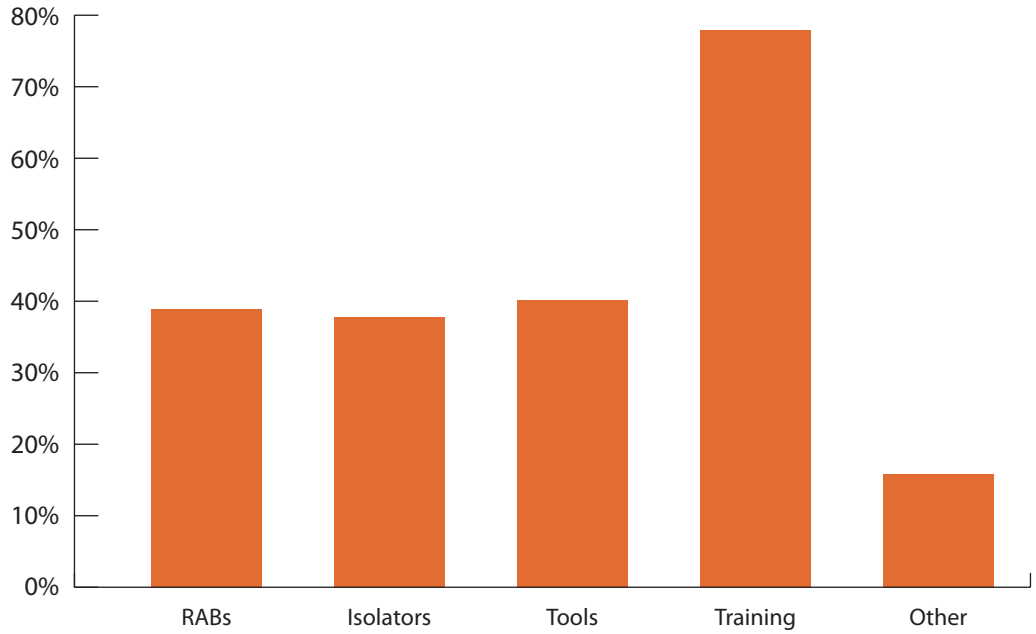
11. If operators are present in your lyophilizer processing area; how do you mitigate the risk of inadvertent contamination of partially stoppered vials during material handling? (please check all that apply)

Answer	Count	%
RABs	32	39.0%
Isolators (or other closed systems)	31	37.8%
Tools	33	40.2%
Training	64	78.0%
Other (please describe)	13	15.9%
	Answered	***82***
	Skipped	***9***

Other (please describe):

- Automated loading machinery.
- Cart under HVAC grade A air.
- Frequent monitoring of operators.
- Full HEPA Filtration Ceiling / Grade A air.
- Grade a HEPA transfer cart.
- Mobile LAFs.
- Movable laminar flow benches.
- No operator in loading area.
- RABS with glove ports.
- The filling and loading of the product in the freeze dryer is done in areas with classification ISO 5.
- Transfer of partially stoppered vials in mobile LAFs.
- Tray covers.
- Vial Discards.



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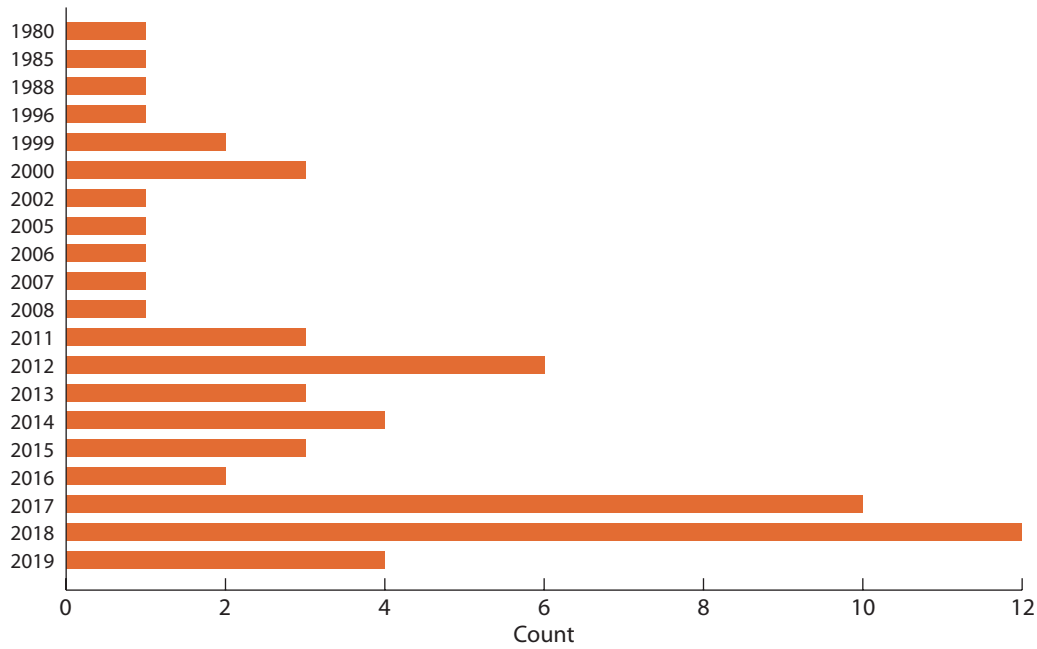
12. When is the last year you had a regulatory filing an NDA where the product was manufactured in a lyophilizer processing area that required personnel presence in the Grade A/B space?

Answer	Count	%
1980	1	1.6%
1985	1	1.6%
1988	1	1.6%
1996	1	1.6%
1999	2	3.3%
2000	3	5.0%
2002	1	1.6%
2005	1	1.6%
2006	1	1.6%
2007	1	1.6%
2008	1	1.6%
2011	3	5.0%
2012	6	9.9%
2013	3	5.0%
2014	4	6.5%
2015	3	5.0%
2016	2	3.3%
2017	10	16.4%
2018	12	19.7%
2019	4	6.5%
	Answered	***61***
	Skipped	***30***



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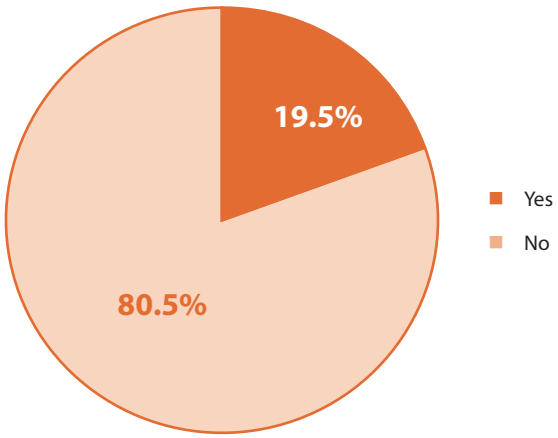
13. Have you received comments/scrutiny from Health Authorities on any NDA filed that included a lyophilizer processing area that required personnel presence in the Grade A/B space?

Answer	Count	%
Yes	16	19.5%
No	66	80.5%
	Answered	***82***
	Skipped	***9***

*If so, please generalize the comment.

Responses:

- Due to aseptic process, the glove port and the type of glove at the rabs was scrutinized and found unsatisfactory.
- General comments on operators moving too quickly in Grade B area when taking micro samples (lyo loading is automated but needs oversight in case of equipment issues).
- Line design to remove people.
- Operators going back and forth from A and B areas.
- Presence of personnel in Grade A space and manipulating partially stoppered product.
- Questions regarding manual loading operations.



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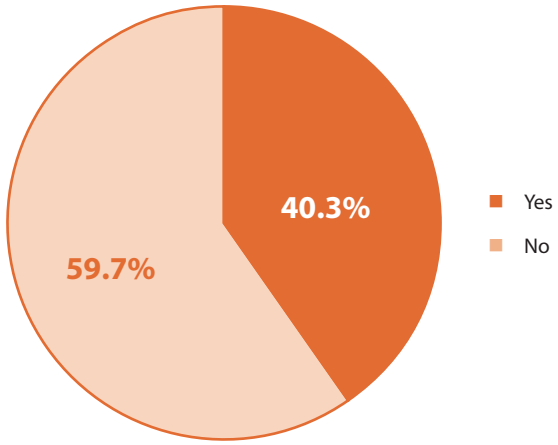
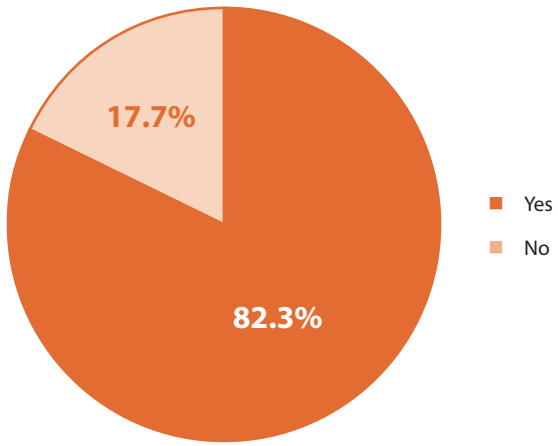
14. Would you have a regulatory filing in 2019 or beyond that utilized a lyophilizer processing area that required operators' presence in the grade A/B space?

Answer	Count	%
Yes	31	40.3%
No	46	59.7%
	Answered	***77***
	Skipped	***14***



15. With regard to future major renovations or new construction, would you exclusively utilize (or consider) fully automated lyophilizer processes?

Answer	Count	%
Yes	65	82.3%
No	14	17.7%
	Answered	***79***
	Skipped	***12***



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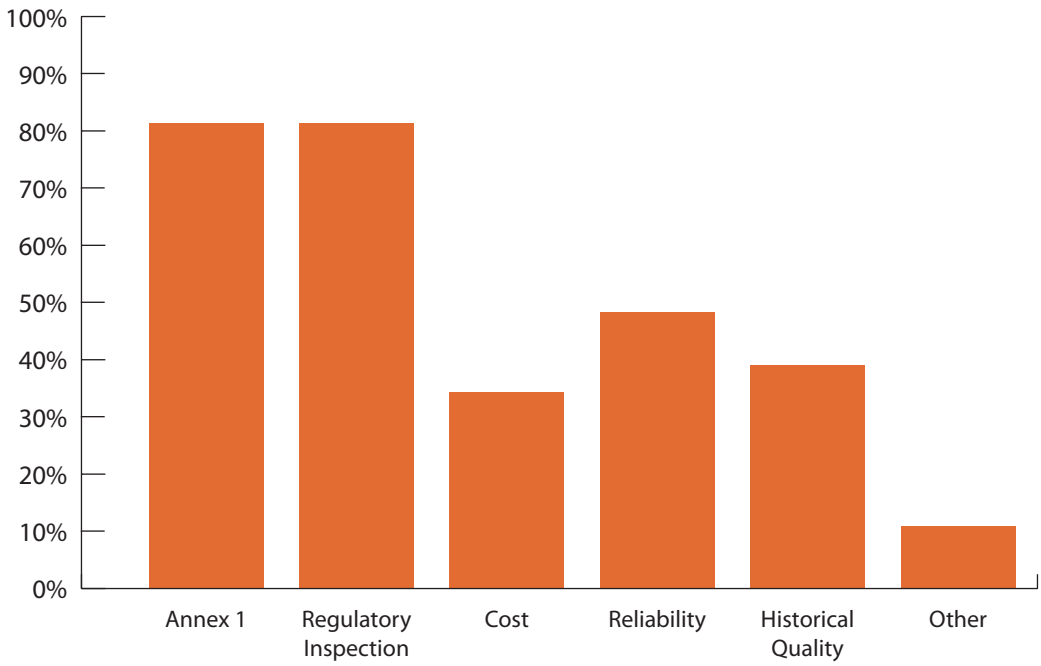
16. If you answered yes, that you would exclusively utilize or consider the use of automated lyophilizer processes for future renovations, what factors have already, or do you expect to, influence that thought process? (please check all that apply)

Answer	Count	%
Annex 1	52	81.3%
Regulatory Inspection	52	81.3%
Cost	22	34.4%
Reliability	31	48.4%
Historical Quality Results with Current Processes	25	39.1%
Other (please define)	7	10.9%
	Answered	***64***
	Skipped	***27***

Other (please define):

- FDA & EMA expectations or the way the expectations are trending.
- Elimination of people from Grade A.
- Expectation of contract givers.
- Operational efficiency, reduction in personnel intervention and increase in probabilistic quality outcome as a result of decreased personnel intervention.
- Operator ease in operation.
- 1. Risk Assessment 2. Minimizing the risk of product contamination with respect to the facility, equipment, personnel's flow and process design for manufacturing Product.
- State of the Art.



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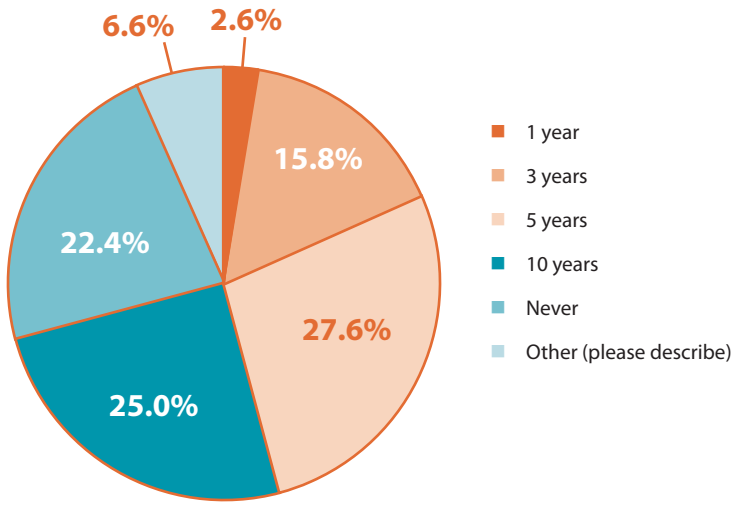
17. Operators in lyophilizer processing areas is clearly an acceptable practice today and no regulations forbid the practice, however, draft Annex 1 encourages the effort to minimize the presence of operators in aseptic areas. When do you believe fully automated lyophilizer processing areas will become the only approvable option for new products?

Answer	Count	%
1 year	2	2.6%
3 years	12	15.8%
5 years	21	27.6%
10 years	19	25.0%
Never	17	22.4%
Other (please describe)	5	6.6%
	Answered	***76***
	Skipped	***15***

Other (please describe):

- I think there will always be facilities that will have operators in the B/A space.
- I think with the pharmaceutical revolution 4.0, the implementation should be accelerated, in addition to the updating of both the standards and the guidelines, it should be said or recommended that everything be automated.
- Risk assessment on product contamination along with nature of products manufactured in the facility.
- Should be by now!
- Up to 10 years, depending on how quickly older installations are upgraded.



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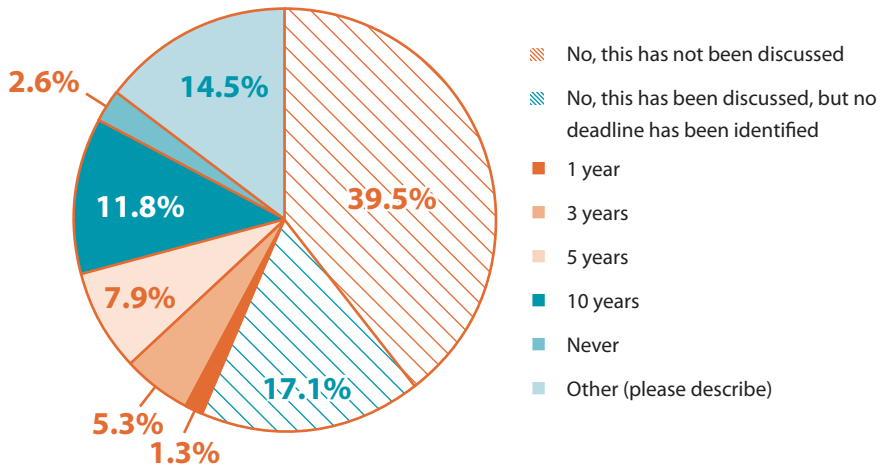
18. Has there been any discussion at your site(s) that all lyophilizer processing areas requiring personnel presence should be de-commissioned? If so, what time frame is being considered?

Answer	Count	%
No, this has not been discussed	30	39.5%
No, this has been discussed, but no deadline has been identified	13	17.1%
1 year	1	1.3%
3 years	4	5.3%
5 years	6	7.9%
10 years	9	11.8%
Never	2	2.6%
Other (please describe)	11	14.5%
	Answered	***76***
	Skipped	***15***

Other (please describe):

- 15-20 years due to new facility upgrades and market complexities.
- All Isolators- no personnel present.
- Already discontinued usage.
- Already fully automated.
- Has been implemented!
- It is already the case.
- No decommissioning required.
- Not discussed for existing lines but new commissioning projects will no longer allow personnel present for lyophilizer loading/ unloading – ALUS only.
- Not relevant as we don't have lyophilization process done in house until the new facility we are currently building. This new facility is built with isolators.
- We do have Isolators therefore n/a.
- We have fully automated system in place with loading, unloading and transfer under isolator.



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19. Please provide any additional feedback with your thoughts about either manual and automated lyophilizer loading or the ways in which lyophilizer loading might be improved now or in the future.

- All commercial product is loaded into lyos using automated systems. Only clinical products are in some cases loaded into smaller lyos without any automated systems.
- Although automated loading systems clearly decrease the contamination risk, we never had a contamination issue related to loading or unloading the lyophilizer. Theoretically it is a great improvement - in practice it just brings more certainty. Unfortunately, GMP authorities force us to perform environmental monitoring in the unmanned loading and unloading areas, so again people have to enter the area.
- Although we utilize fully automated loading systems within a RABS enclosure, operators are still required to be present and conduct aseptic interventions. These questions were worried in a manner that did not accommodate automated loading with the presence of operators in a Grade A/B area.
- At least for small scale commercial and clinical lyo manufacturing manual loading will prevail for the next years in our company.
- Automated lines have many troubles.
- Automated loading is by far more reliable.
- Automated lyophilizer is an achievable goal, however, at the moment the cost of improving existing line is too much for us to consider, as there are no regulations forbidding the use of personnel in aseptic grade a/b areas yet.
- Automated lyophilizer loading must be approved immediately.
- Automatic loading is the best option to achieve desired SAL and safeguard product quality.
- Continuous barrier between filling and FD loading is now approved technology so we need to move to that increased protection of our products. But this cannot be done within a short time delay. 3 or even more years will be necessary. A short registration process should be proposed for all *cies* to be able to switch to that new design.
- For new facilities there should be no consideration. Just build with automated loading systems and use isolator technology to separate human and product. For older facilities a change to automated loading and use of isolators might not be an acceptable solution. This is the reason I think it will be very long before it will be a requirement for new products in existing facilities.
- I believe automated loading is best practice and gives higher sterility assurance (albeit hard to measure that)
- I'm worried that we're still discussing this!
- Its proven here that automated process is the way to minimize contamination and maintain controls.
- Lyophilizer loading and unloading of half stoppered vials should be under closed Systems without any manual intervention till the vials are capped.
- Manual loading/unloading is hopefully going to be replaced with automatic loading and unloading.
- May have to stop producing lyophilized products.

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- Modelling and automation.
- Our manufacturing is all done externally so you may want to ask the question up front.
- Perform Quality risk assessment on product contamination with respect to the facility, equipment, personnel's flow and process design for manufacturing the products at site. Usage of isolation, decontamination, containment, dedication and disposable techniques during LYOPHILIZATION process design (QbD).
- Risk based approach should be considered. Different approach of continuous manufacturing vs risk concern.
- The field of robotics and gloveless isolators has advanced very much and currently, such systems are being commissed. I think with this paradigm shift, manual lyo loading will also be retought. Due to this, regulators will not accept manual loading of lyophilizers in the future and will force the manufacturers to re-think their loading process.
- The main issue with ALUS in my company is that it would require the complete reconstruction/ renovation of quite old filling lines which significantly increases the cost.
- We have product dedicated lyophilizer, but the problem comes when the fill volume or dose volume changes along with vial size and stopper design. At this point of time considering a fully automated system requires a risk-reliability-continuous improvement based approach. Until the ANNEX-1 is finalized we are uncertain about the final expectations. Furthermore we might deal with multiple vendors and consultants for complete automation of existing lyophizer. The criticality remains on the down time during such an upgrade which might seriously impact out product supply.
- Well, with regard to manual loading, I think that filling interventions or any type of intervention that is carried out through risk analysis should be evaluated and that these interventions are retained in the media fill.
- Working on hydraulic lift to allow use of push tools to separate operator by distance. Not sure is this will be accepted yet.

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CONCLUSION

The *2019 Sterile Lyophilized Drug Product Loading Survey* was intended for individuals that manufacture, develop, or contract for the manufacture of lyophilized drug products. As expected, companies continue to have some lyophilization operations that require manual intervention during the loading phase. The aggregated information generally indicates that these operations have some inherent risk. Although regulatory criticism has not always been received, the regulatory attitudes remain an important driver for change in the future. Nevertheless, there is difficulty with excluding the practice within existing facilities and installations and time will be required to adopt substantive changes in this area.

The survey will support PDA, industry and regulators in understanding the current state of the industry, as well as getting insights to the discussions that may be occurring at the site level to enhance sterility assurance for the critical lyophilizer loading operations.

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